

Learning Objectives

- Relate viscosity to strength of IMFs
- Understand the process of vaporization and predict relative vapor pressures, boiling points, and rate of evaporation for liquids
- Understand the relationship between vapor pressure and atmospheric pressure and how it relates to boiling point.

Viscosity

- Viscosity: fluid's resistance to flow. *thick / thin*
- higher IMFs, more viscous the fluid.

TABLE 12.5 ■ Viscosity of Several Hydrocarbons at 20 °C

Hydrocarbon	Molar Mass (g/mol)	Formula	Viscosity (cP)
<i>n</i> -Pentane	72.15	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	0.240
<i>n</i> -Hexane	86.17	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	0.326
<i>n</i> -Heptane	100.2	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	0.409
<i>n</i> -Octane	114.2	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	0.542
<i>n</i> -Nonane	128.3	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	0.711

viscosity increase

Dispersion

- longer length of the molecule (molecular entanglement), more viscous the fluid.
- lower temperature, more viscous the fluid.

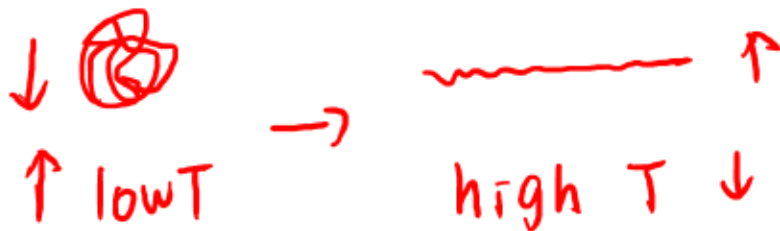


TABLE 12.6 ■ Viscosity of Liquid Water at Several Temperatures

Temperature (°C)	Viscosity (cP)
20	1.002
40	0.653
60	0.467
80	0.355
100	0.282

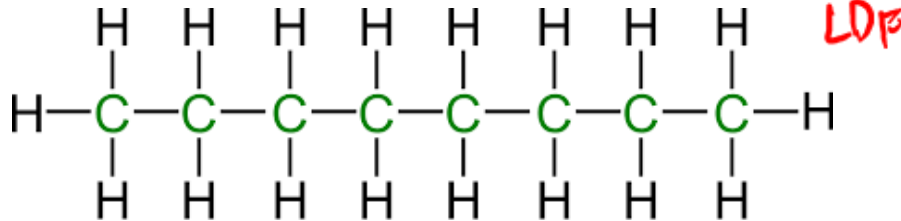
increase

decrease

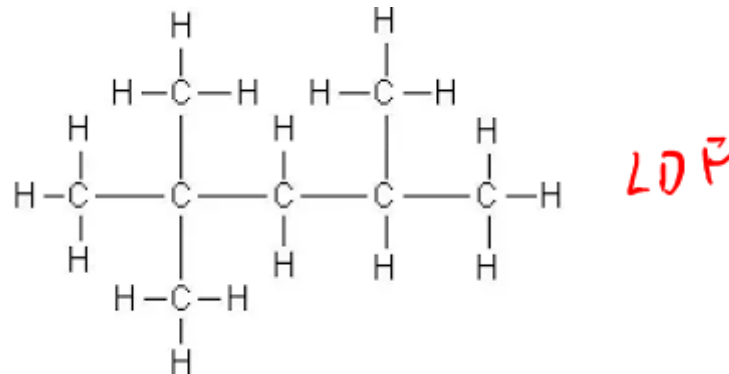
Your Turn!

- At the same temperature, which species has a highest viscosity?

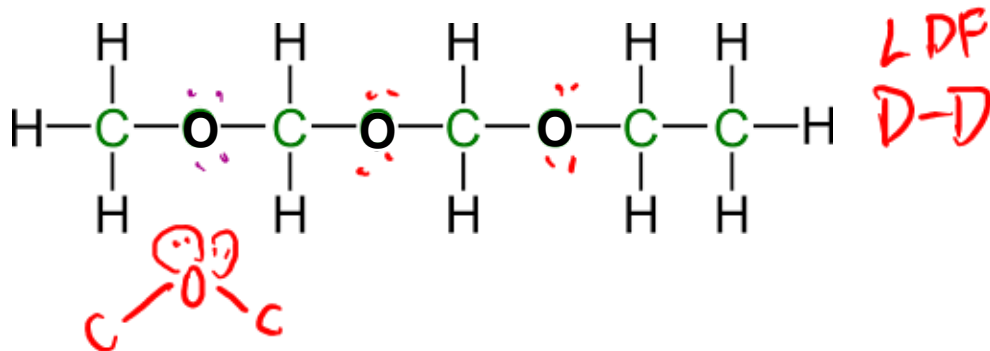
A.



B.

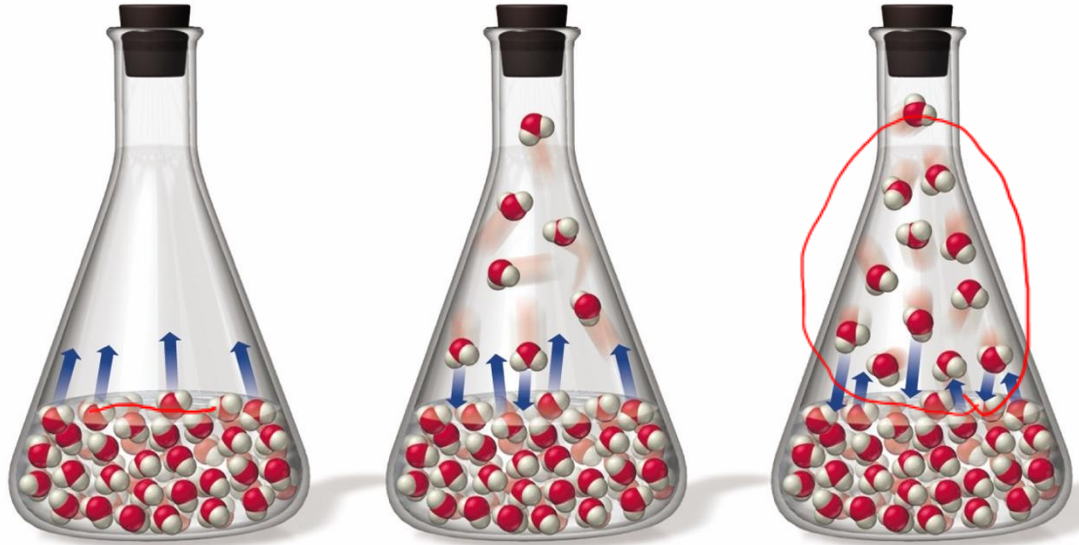
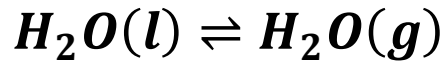


C.

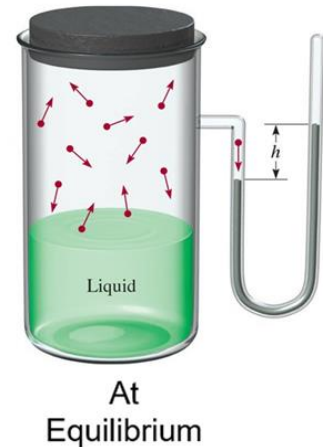
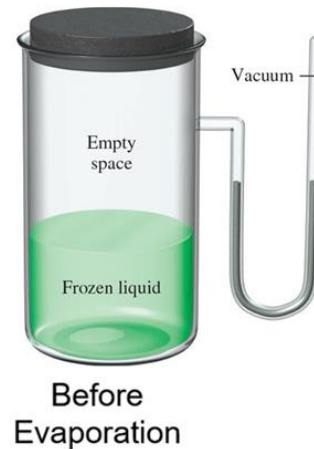
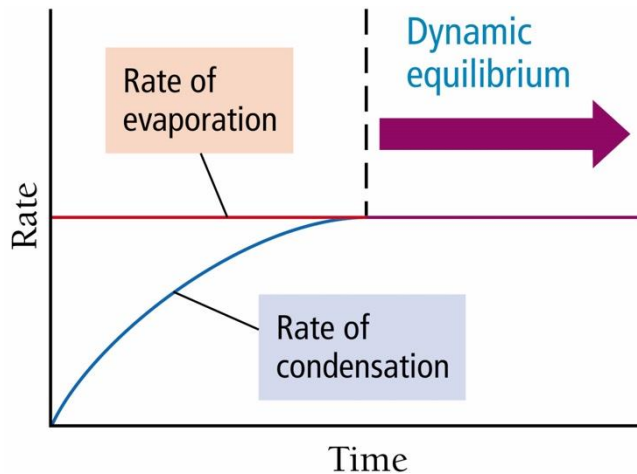


Vapor Pressure of Pure Liquids

- Vapor pressure: pressure contributed by a vaporized solid/liquid at equilibrium



- At the equilibrium, the rate of evaporation = the rate of condensation



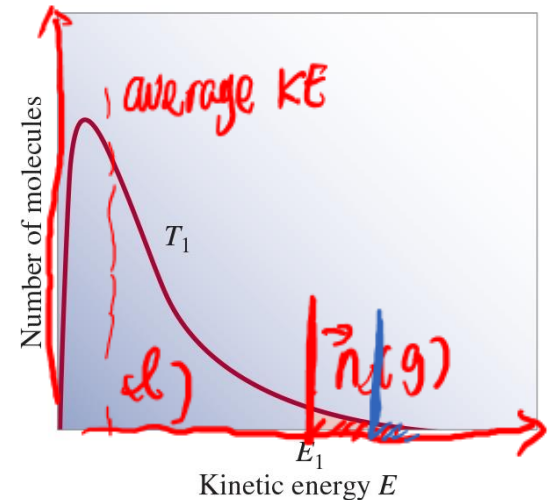
Escape Kinetic Energy

The escape kinetic energy required for molecules escape into the gas phase

- For liquid/solid:

Escape KE > Average KE

- more escaped molecules, higher vapor pressure

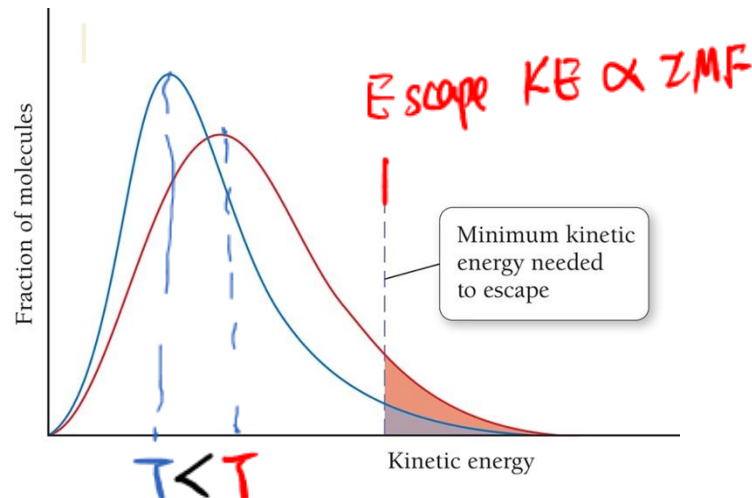


At the same temperature:

higher IMF, higher escape kinetic energy, lower vapor pressure

Same substance (same escape KE):

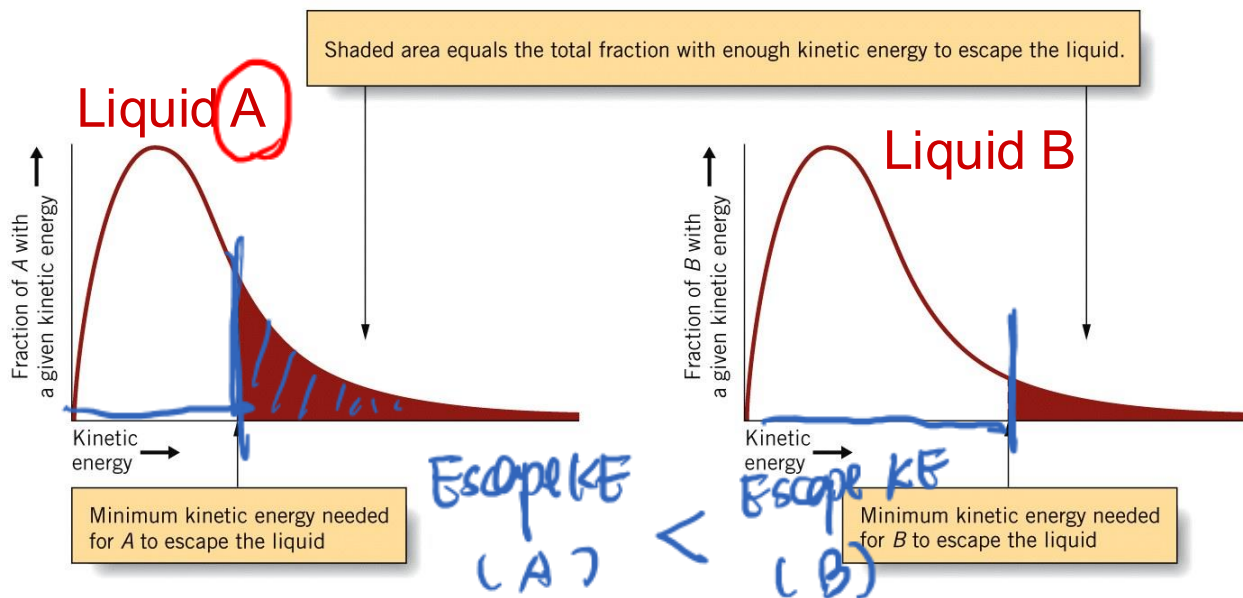
higher temperature, higher vapor pressure



Your Turn!

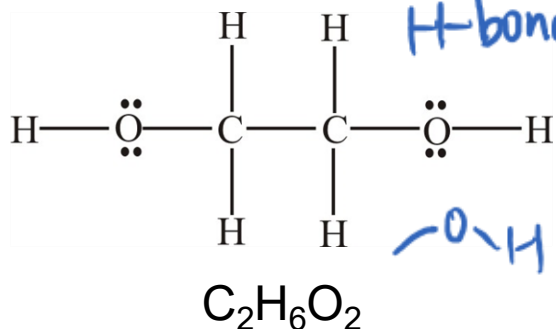
There are two liquids at the same temperature. One of them is $\text{C}_2\text{H}_6\text{O}_2$ and the other is $\text{C}_3\text{H}_6\text{O}$.

Q1. At the same temperature, which liquid will have a higher vapor pressure?

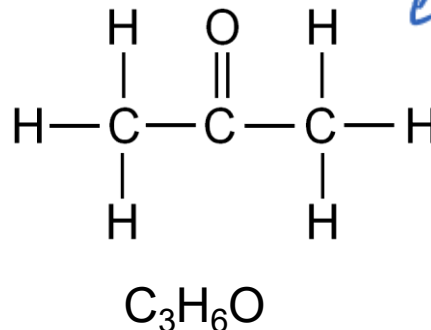


Q2. Given the structures, which one is more likely to be A?

1)



2)



Announcements 3/9

Assignments that are due soon

- Pre-Class Assignment Week 6-W

Office Hour

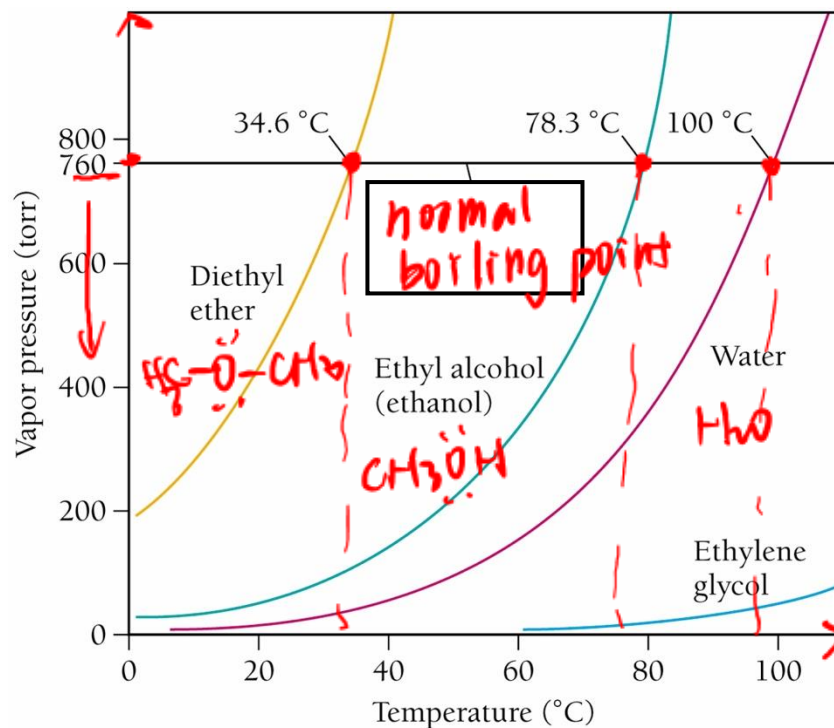
- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:~~30~~-1:~~30~~ PM at the TLC (Demille 160P)

Boiling Point

- The boiling point is the temperature at which the equilibrium vapor pressure of a liquid is equal to the external pressure.
- “Bubbles” are liquid vapor (NOT air!)



- Normal boiling point** is the temperature at which the vapor pressure of a substance reaches 1 atm
760 torr



Your Turn!

The elevation of the Mount Everest, Tibetan Plateau is about 29029 ft, and the atmosphere pressure is about 0.33 atm/250 torr.



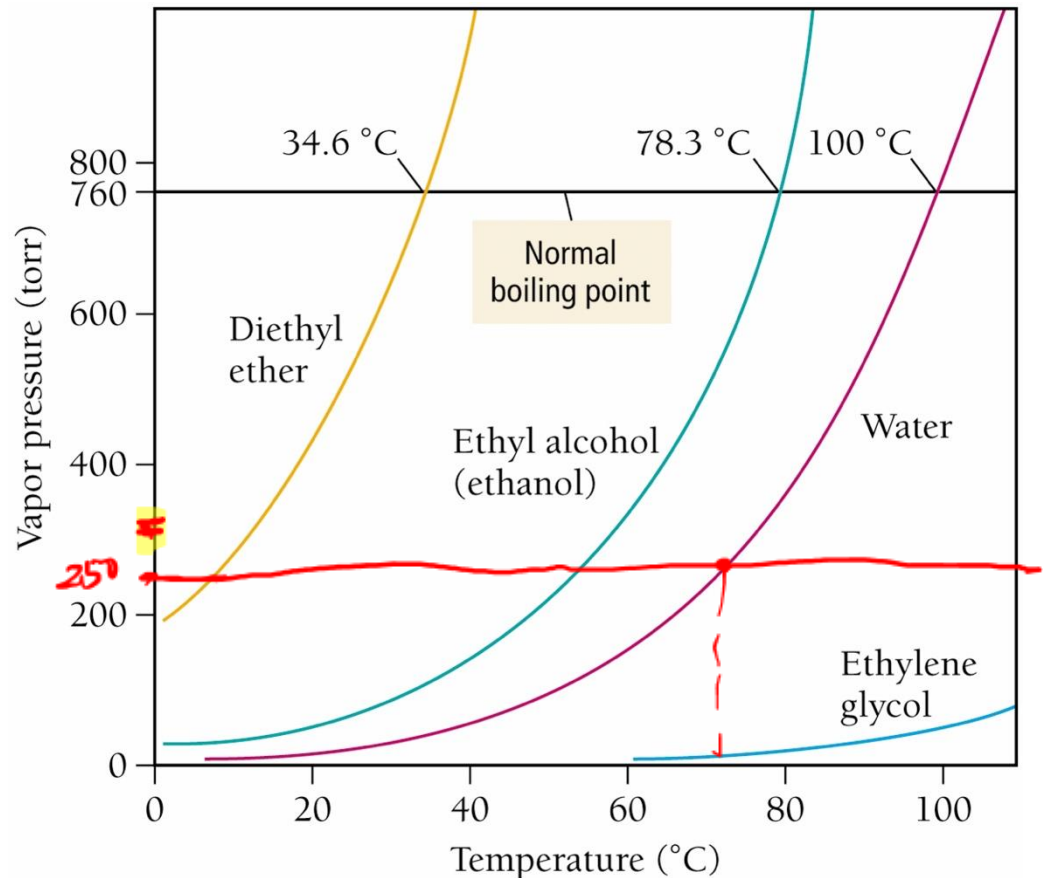
In which temperature will the water boil at Mount Everest?

A. 100 °C

B. 70 °C

C. 50 °C

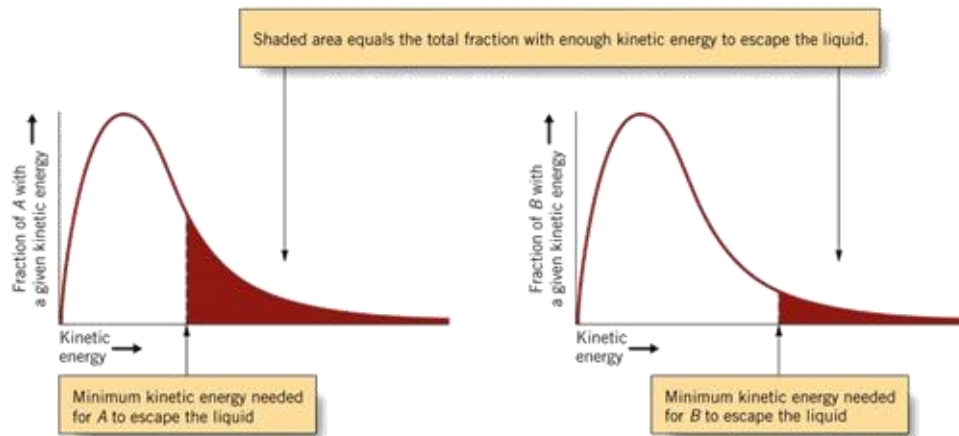
D. 0 °C



Boiling Point

- The boiling point is the temperature at which the equilibrium vapor pressure of a liquid is equal to the external pressure.

- higher IMFs, higher boiling point

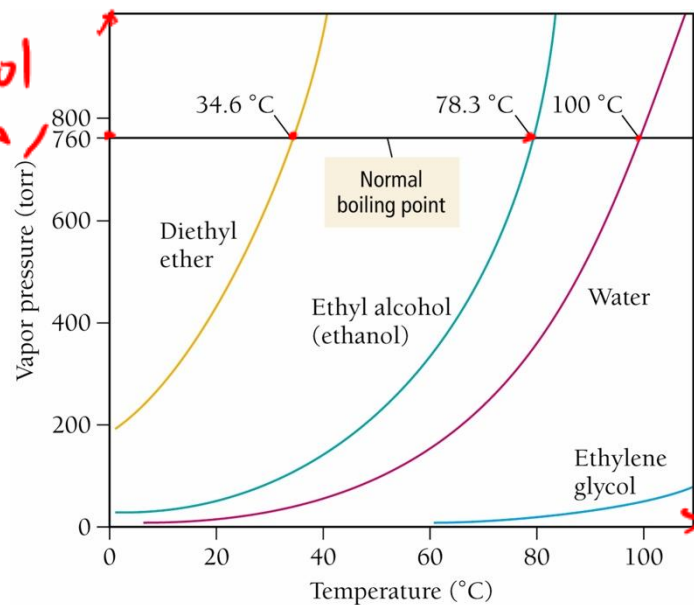


- ΔH_{vap} : Enthalpy of evaporation

Heat required to evaporate 1 mol of a liquid to gas

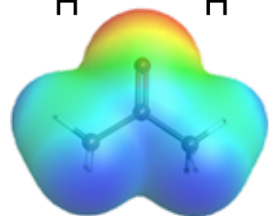
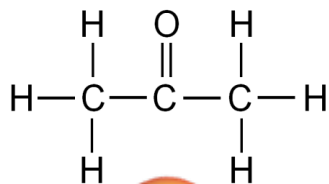


- higher IMFs, higher ΔH_{vap}



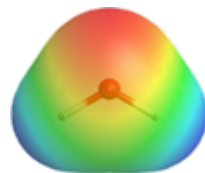
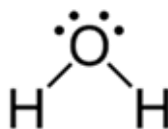
Your Turn!

Identify the intermolecular forces in the three compounds and rank them by...



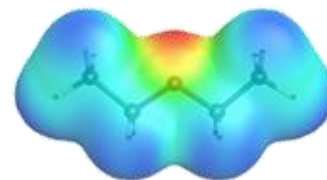
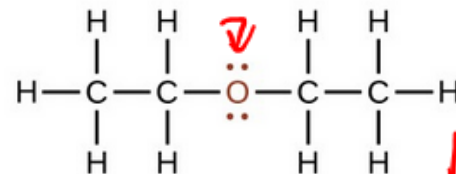
1) Acetone

LDF
D-D



2) Water

LDF
D-D
H-bond



3) Diethyl Ether

LDF
D-D



- A. Order of strength of IMFs $\uparrow 2) > 1) > 3)$
- B. Order of boiling points $\uparrow 2) > 1) > 3)$
- C. Order of vapor pressure in a closed flask at 25 °C $\downarrow 3) > 1) > 2)$
- D. Order of Heat of Vaporization $\Delta H_{\text{vap}} \uparrow 2) > 1) > 3)$